Liver Tumor Segmentation Using Deep Learning: A Comprehensive Survey and Comparative Analysis

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***Abstract*—Liver cancer is among the most lethal malignancies worldwide, with hepatocellular carcinoma (HCC) accounting for approximately 75–85% of primary liver cancers and an estimated 758,000 deaths annually according to GLOBOCAN 2022. Accurate and reproducible delineation of liver tumors from Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) scans is a clinical prerequisite for treatment planning, surgical navigation, and longitudinal disease monitoring. Manual segmentation by expert radiologists is time-consuming (20–60 minutes per volume), subject to significant inter-observer variability (10–20% Dice score), and impractical at the scale of clinical reporting. This paper presents a comprehensive survey and comparative analysis of deep learning methods for liver tumor segmentation spanning 2020–2024, covering over 20 distinct architectures organized into five paradigmatic categories: (1) encoder-decoder networks and U-Net variants, (2) attention-augmented architectures, (3) Vision Transformer-based models, (4) Generative Adversarial Network (GAN)-based augmentation and synthesis frameworks, and (5) semi-supervised and federated learning approaches. A systematic comparison is conducted on the Liver Tumor Segmentation Benchmark (LiTS) and 3DIRCADb datasets using standard metrics including Dice Coefficient (DC), Intersection over Union (IoU), Average Surface Distance (ASD), and Hausdorff Distance (HD). Results show a clear performance trajectory: tumor Dice scores improved from approximately 0.74 in CNN-based methods (2020) to 0.83–0.94 in Transformer and foundation model-based approaches (2023–2024). Remaining open challenges including class imbalance, domain shift, annotation scarcity, and clinical translation requirements are discussed, and future research directions including federated learning, self-supervised pretraining, and real-time deployment are identified.**

***Index Terms*—liver tumor segmentation, deep learning, U-Net, Vision Transformer, convolutional neural network, attention mechanism, GAN, LiTS dataset, Dice coefficient, federated learning, semi-supervised learning, medical image analysis, computer-aided diagnosis, hepatocellular carcinoma**

I. Introduction

Liver cancer ranks sixth in global cancer incidence and third in cancer mortality, with an estimated 865,000 new cases and 758,000 deaths per year (GLOBOCAN 2022) [1]. The five-year overall survival rate remains below 20%, highlighting the critical importance of early and accurate diagnosis. Hepatocellular carcinoma (HCC), constituting 75–85% of primary liver tumors, arises from hepatocytes and is strongly associated with chronic hepatitis B/C infection, liver cirrhosis, and non-alcoholic steatohepatitis (NASH).

Accurate liver tumor segmentation from medical imaging is a foundational step in clinical management. Tumor volume, shape, and spatial relationship to hepatic vasculature directly inform decisions regarding surgical resectability, locoregional therapy eligibility (transarterial chemoembolization, radiofrequency ablation, stereotactic body radiotherapy), and response assessment per RECIST criteria [2]. Manual segmentation requires 20–60 minutes of expert effort per CT volume and exhibits inter-observer Dice variability of 10–20%, making it impractical at clinical scale.

Automated segmentation using machine learning and, more recently, deep learning has transformed this domain. The publication of the U-Net architecture [3] and the establishment of the Liver Tumor Segmentation Benchmark (LiTS) [4] in 2017 galvanized community effort. Subsequent architectural innovations—residual connections, dense connections, attention gates, and Vision Transformers—have driven progressive performance improvements.

The principal challenges in liver tumor segmentation are:

- *Extreme class imbalance*: tumor voxels constitute $< 3\%$ of total CT volume, frequently $< 0.5\%$ for small early-stage lesions.
- *Morphological variability*: tumor diameters range from 4 mm to 187 mm in the LiTS dataset, spanning sub-centimeter nodules to massive multinodular masses.
- *Low contrast*: hypovascular tumors may be isodense to surrounding parenchyma in non-contrast or early-phase CT.
- *Annotation scarcity*: the LiTS training set contains only 131 annotated CT volumes, orders of magnitude smaller than natural image datasets used to pre-train deep models.
- *Domain shift*: models trained at one institution typically degrade 5–10% in Dice score when deployed at another due to scanner and protocol variability.

This paper makes the following contributions:

- A systematic review of 20+ deep learning architectures for liver tumor segmentation (2020–2024), organized by methodology.
- Quantitative comparison of methods on LiTS and 3DIRCADb using Dice, IoU, ASD, and HD metrics.

- Analysis of loss functions, preprocessing strategies, and evaluation protocols.
- Identification of open challenges and future research directions including federated learning and foundation models.

II. Background and Related Work

A. Medical Imaging for Liver Cancer

Computed Tomography (CT) is the gold standard for liver tumor staging and automated segmentation research. Liver parenchyma appears at 50–70 HU; hypovascular tumors at 20–40 HU; hypervascular HCC at 80–120 HU in the arterial phase. Standard preprocessing applies HU windowing to $[-200, 250]$ (liver window) followed by z-score normalization. Magnetic Resonance Imaging (MRI) offers superior soft-tissue contrast, particularly with hepato-specific contrast agents (gadoxetic acid), but lacks a standardized intensity scale.

B. Classical Segmentation Methods

Pre-deep learning approaches employed handcrafted features—Grey-Level Co-occurrence Matrix (GLCM), Histogram of Oriented Gradients (HOG), Local Binary Patterns—combined with Support Vector Machines (SVM) or Random Forests. Devi and Seenivasagam [5] achieved 82% accuracy using SVM with GLCM features on private datasets. These methods suffer from sensitivity to parameter tuning and cannot learn hierarchical spatial representations critical for complex tumor morphology.

C. The LiTS Benchmark

The Liver Tumor Segmentation Benchmark (LiTS) [4] contains 201 abdominal CT volumes (131 training, 70 test) from seven hospitals with diverse scanner models and protocols. Ground truth annotations were performed by expert radiologists. The benchmark has tracked continuous improvement since 2017, with the mean tumor Dice score increasing from approximately 0.52 (2017 top submission) to 0.78 (2022 top submission), demonstrating the benchmark's effectiveness as a community driver.

III. Methodology Overview

A. Problem Formulation

Let $\mathcal{V} \in \mathbb{R}^{H\times W\times D}$ denote a 3D CT volume with $H \times W$ in-plane resolution and D axial slices. The goal is to learn a mapping $f_\theta : \mathcal{V} \to \mathcal{M}$, where $\mathcal{M} \in \{0, 1\}^{H\times W\times D}$ is the voxel-wise binary segmentation mask with label 1 for tumor tissue and 0 for background. The mapping f_θ is parameterized by neural network weights θ trained to minimize a segmentation loss $\mathcal{L}$.

B. Preprocessing Pipeline

HU Windowing: CT intensities are clipped to $[-200, 250]$ HU, isolating hepatic soft-tissue relevant for tumor characterization and discarding irrelevant bone and air intensities.

Spatial Resampling: Volumes are resampled to isotropic spacing ($1.0\times1.0\times1.0\,\mathrm{mm}^3$ for 3D networks) using third-order spline interpolation for images and nearest-neighbor for masks.

Intensity Normalization: After windowing, voxel intensities are z-score normalized ($\mu = 0$, $\sigma = 1$) per volume to correct for residual inter-scanner variability.

Data Augmentation: Random horizontal/vertical flipping, rotation ($\pm30°$), zoom (0.8–1.2×), elastic deformation, and intensity jitter ($\pm20\%$) are applied. Tumor-biased patch sampling ensures each training patch contains at least one tumor voxel with probability $p = 0.5$.

C. Evaluation Metrics

The following metrics are used for performance evaluation:

Dice Coefficient (DC): Primary metric measuring volumetric overlap:

$$\mathrm{DC} = \frac{2|X \cap Y|}{|X| + |Y|}, \tag{1}$$

where X and Y denote predicted and ground truth masks respectively.

Intersection over Union (IoU):

$$\mathrm{IoU} = \frac{|X \cap Y|}{|X \cup Y|} = \frac{\mathrm{DC}}{2 - \mathrm{DC}}. \tag{2}$$

Average Surface Distance (ASD): Mean distance (mm) between predicted and ground truth surfaces, measuring boundary accuracy.

Hausdorff Distance (HD): Maximum of minimum distances between surface points, sensitive to outlier boundary deviations.

Sensitivity (Recall): $\mathrm{Sen} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$, the most clinically critical metric for tumor detection.

D. Loss Functions

Dice Loss: $\mathcal{L}_{\mathrm{Dice}} = 1 - \mathrm{DC}$. Directly optimizes the primary evaluation metric; robust to class imbalance.

Focal Loss [6]: Down-weights easy negative voxels: $\mathcal{L}_{\mathrm{Focal}} = -\alpha_t(1-p_t)^\gamma \log p_t$, with $\gamma = 2$ reducing dominance of background voxels.

Tversky Loss: Asymmetric extension of Dice:

$$\mathcal{L}_{\mathrm{TV}} = 1 - \frac{\mathrm{TP}}{\mathrm{TP} + \alpha \cdot \mathrm{FP} + \beta \cdot \mathrm{FN}}, \tag{3}$$

with $\beta > 0.5$ prioritizing recall for small tumor detection.

Compound Loss: Most architectures combine Dice and Focal: $\mathcal{L} = \lambda_1\mathcal{L}_{\mathrm{Dice}} + \lambda_2\mathcal{L}_{\mathrm{Focal}}$.

IV. Deep Learning Architectures for Liver Tumor Segmentation

A. U-Net and Encoder–Decoder Variants

The U-Net [3] established the encoder-decoder paradigm with skip connections that has dominated medical image segmentation. The encoder progressively reduces spatial resolution while increasing feature depth; the decoder mirrors this structure with transposed convolutions and skip connections

concatenating encoder features at matched resolutions to recover spatial detail.

V-Net (Milletari et al., 2016) [7] extended U-Net to 3D volumetric inputs with 3D convolutions and introduced the Dice loss as the training objective. 3D processing captures inter-slice anatomical continuity naturally but requires patch-based training due to memory constraints.

H-DenseUNet (Li et al., 2018) [8] combined 2D DenseUNet (processing individual axial slices) with 3D DenseUNet (capturing volumetric context) in a hybrid architecture, achieving a tumor Dice of 0.780 on LiTS and setting the state of the art at the time.

UNet++ (Zhou et al., 2018) [9] redesigned skip connections into nested dense paths, eliminating the semantic gap between encoder and decoder representations. Intermediate nodes in each nested path perform additional feature fusion with progressively better encoder-decoder alignment.

ResUNet (Rahman et al., 2022) [10] integrated residual blocks with identity shortcuts into the U-Net framework, mitigating vanishing gradients in deep networks and achieving a tumor Dice of 0.792 on LiTS. Residual learning is particularly effective for small tumor detection due to improved gradient flow to early encoder layers.

nnU-Net (Isensee et al., 2021) [11] introduced a self-configuring framework that automatically determines preprocessing, network topology, and training protocol based on dataset fingerprints. Without dataset-specific architectural modifications, nnU-Net achieved a tumor Dice of 0.810, outperforming many manually designed architectures and establishing the critical importance of systematic hyperparameter optimization alongside architectural novelty.

AIM-Unet (Özcan et al., 2023) [12] hybridized U-Net with Inception modules, applying parallel convolutions at kernel sizes $\{1\times1, 3\times3, 5\times5\}$ within each encoder block to capture multi-scale features simultaneously. Validated on four datasets (CHAOS, LiTS, 3DIRCADb, private), AIM-Unet achieved approximately 0.800 tumor Dice on LiTS.

EREs-UNet++ (Li et al., 2023) [13] extended the nested UNet++ with High-Efficiency Channel Attention (HECA)—a 1D depthwise convolution approximating standard channel SE blocks at lower computational cost—achieving a tumor Dice of 0.779 with improved sensitivity for sub-centimeter lesions.

B. Attention-Augmented Architectures

Attention U-Net (Oktay et al., 2018) [14] introduced additive attention gates at each skip connection. The attention gate computes:

$$\alpha = \sigma\big(W_\alpha \cdot \mathrm{ReLU}(W_g g + W_x x + b)\big), \tag{4}$$

producing a soft spatial mask $\hat{x} = \alpha \odot x$ that suppresses background activations and highlights tumor-relevant features.

RA-UNet (Jin et al., 2020) [15] combined residual learning with attention gates at multiple encoder-decoder levels, achieving a liver Dice of 0.960 and tumor Dice of 0.741 on LiTS.

RMAU-Net (Jiang et al., 2023) [16] incorporated residual multi-scale dilated convolutions at rates $\{1, 2, 4\}$ with attention gates at every skip connection, enabling simultaneous extraction of local texture and global structural features for variable-size tumors. Tumor Dice of 0.762 was reported on LiTS with improved boundary precision.

CPAD-Net (Wang et al., 2023) [17] employed a dual attention module combining channel-wise Squeeze-and-Excitation (SE) with spatial attention, applied to parallel dilated convolution branches at rates $\{1, 2, 4, 8\}$, enlarging the effective receptive field while focusing computational capacity on diagnostically relevant regions.

MDAUnet (Saumiya and Franklin, 2024) [18] introduced a multi-task deformable attention U-Net for simultaneous liver tumor segmentation and classification. Deformable convolutions learn spatially adaptive sampling positions, conformally adapting to irregular tumor shapes without geometric assumptions. Multi-task training jointly optimized segmentation and classification losses, improving both tasks through shared spatial feature representations.

C. Vision Transformer-Based Models

Vision Transformers model global dependencies via multi-head self-attention (MSA):

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{5}$$

enabling global context modeling from the first network layer—a fundamental capability unavailable to locally-connected CNNs.

UNETR (Hatamizadeh et al., 2022) [19] replaced the CNN encoder with a pure Vision Transformer (ViT), partitioning 3D volumes into $16 \times 16 \times 16$ voxel patches and processing the resulting sequence through 12 transformer layers. Feature representations extracted at depths $\{3, 6, 9, 12\}$ feed CNN decoder branches with skip connections, combining global context with local spatial recovery. Tumor Dice of 0.772 on the Medical Segmentation Decathlon (MSD).

Swin-UNet (Cao et al., 2022) [20] constructed a pure transformer-based U-Net using hierarchical Swin Transformer blocks. Shifted-window attention limits complexity to $\mathcal{O}(N)$ in image resolution, enabling application to high-resolution 3D CT patches. Patch merging (encoder) and patch expanding (decoder) create a structure directly analogous to convolutional U-Net downsampling and upsampling.

nnFormer (Zhou et al., 2023) [21] interleaved local window attention and global volumetric attention, balancing fine-grained boundary detail with whole-volume context, achieving a tumor Dice of 0.778 on LiTS.

TD-Net (Di et al., 2023) [22] employed a dual-path hybrid combining a transformer attention path with a dilated CNN path processing identical features in parallel, achieving liver Dice of 0.962 and tumor Dice of 0.763 on LiTS.

FUF-TransUNet (Song et al., 2024) [23] augmented the TransUNet framework with two modules: a Dense-Scale (DS) module aggregating multi-scale encoder features via dense

connections, and a Stepwise Feature Fusion (SFF) module adaptively weighting and integrating features at each decoder step. Achieved the highest transformer-based performance at publication: tumor Dice of 0.834 on LiTS and 0.898 on 3DIRCADb.

G-UNETR++ (2026) [24] extended UNETR by incorporating gradient magnitude images as an additional input channel, providing explicit boundary awareness to the transformer encoder. Gradient maps computed via Sobel operators highlight intensity discontinuities at tumor-parenchyma interfaces. Achieved a tumor Dice of 0.844 on LiTS and 0.832 on 3DIRCADb.

D. GAN-Based Augmentation and Synthesis

MedGAN (Guo et al., 2023) [25] employed progressive training—beginning at low resolution and scaling to full resolution— for multi-phase CT liver tumor synthesis. The conditional GAN objective:

$$\min_{G} \max_{D} \; \mathbb{E}_{x,y}\big[\log D(x|y)\big] + \mathbb{E}_{z,y}\big[\log(1 - D(G(z|y)|y))\big], \tag{6}$$

conditions generation on target phase labels, enabling targeted augmentation across imaging phases underrepresented in available datasets.

MA-CGAN (Liao et al., 2024) [26] integrated multi-axis attention with a conditional GAN within a unified segmentation framework. The multi-axis attention computes attention weights along each spatial axis independently—$\mathcal{O}(H + W + D)$ complexity vs. $\mathcal{O}(HWD)$ for full 3D attention—and the cGAN loss regularizes predictions toward anatomically realistic boundaries. HD was reduced by 18% and recall for lesions $<$2 cm improved by 6.2% compared to baseline U-Net.

E. Semi-Supervised and Federated Learning

Label-Free Segmentation (Hu et al., CVPR 2023) [27] demonstrated that high-quality pseudo-labels generated by the Segment Anything Model (SAM) could train competitive supervised segmentation networks, achieving tumor Dice of 0.760 with zero manual annotations.

Multi-View SSL (npj Digital Medicine, 2025) [28] coupled a semi-supervised segmentation backbone with a foundation model-guided regularizer and multi-view collaborative learning (axial, sagittal, coronal). Using only 20 labeled cases, the framework achieved liver Dice of 83.79% and tumor Dice of 60.08%, outperforming prior semi-supervised methods.

FedLiverNet (Scientific Reports, 2025) [29] presented a federated liver and tumor segmentation system with differential privacy aggregation and clustered federated learning, training across simulated institutional partitions of LiTS without sharing patient data. Performance was within 5% of centralized training while satisfying GDPR/HIPAA data locality requirements.

SAM + GAN (Demir et al., 2023) [30] combined the Segment Anything Model with GAN-based domain adaptation for CT, achieving the highest reported LiTS tumor Dice of 0.943, recall of 0.952, and accuracy of 0.938, demonstrating the exceptional pretraining advantage of foundation models.

V. Experimental Results and Comparison

A. Datasets

Table I summarizes the benchmark datasets used across the reviewed literature.

TABLE I
Benchmark Datasets for Liver and Tumor Segmentation

Dataset	Volumes	Structures	Modality
LiTS17 [4]	131/70	Liver, Tumor	CT (multi-center)
3DIRCADb [31]	20	Liver, Tumor, Vessels	CT
CHAOS [32]	40/120	Liver (no tumor GT)	CT + MRI
MSD Task 08	443/140	HCC Tumors	CT (portal-venous)
TCGA-LIHC	Variable	HCC	CT + MRI

B. Quantitative Comparison

Table II reports liver and tumor segmentation performance for all reviewed methods on the LiTS dataset. Results are drawn from primary publications and official CodaLab submissions.

TABLE II
Performance Comparison on LiTS Dataset. DC = Dice Coefficient (↑ higher is better). ASD = Average Surface Distance in mm (↓ lower is better). N/A = not reported.

Method	Year	Liver DC	Tumor DC	ASD
Cascade FCN [33]	2017	0.946	0.675	N/A
H-DenseUNet [8]	2018	0.961	0.780	N/A
RA-UNet [15]	2020	0.960	0.741	14.8
U-n-Net [34]	2020	0.951	0.741	21.3
SegNet (Mod.) [35]	2020	0.943	0.790	18.4
SAA-Net [36]	2022	0.963	0.756	9.1
ResUNet [10]	2022	0.958	0.792	8.7
nnU-Net [11]	2021	0.963	0.810	7.9
UNETR [19]	2022	0.952	0.772	11.2
AIM-Unet [12]	2023	0.964	0.800	8.6
EREs-UNet++ [13]	2023	0.963	0.779	8.2
RMAU-Net [16]	2023	0.961	0.762	8.9
TD-Net [22]	2023	0.962	0.763	8.4
CPAD-Net [17]	2023	0.960	0.768	9.0
MDAUnet [18]	2024	0.965	0.783	8.0
MA-CGAN [26]	2024	0.965	0.771	7.8
FUF-TransUNet [23]	2024	0.968	0.834	6.9
G-UNETR++ [24]	2026	0.971	0.844	5.8
SAM + GAN [30]	2023	0.976	0.943	5.1

C. Analysis of Results

Several trends emerge from Table II.

Liver vs. Tumor: Liver segmentation Dice saturates between 0.94 and 0.98 across all reviewed methods, indicating this subtask is near-solved on LiTS. Tumor segmentation shows far greater variation (0.675–0.943), confirming it as the dominant remaining challenge.

Architectural progression: Tumor Dice progressed from 0.675 (Cascade FCN, 2017) through 0.74–0.79 (CNN and attention U-Net methods, 2020–2022) to 0.83–0.84 (transformer-based hybrids, 2023–2024). The SAM + GAN

result (0.943) demonstrates the exceptional pretraining advantage of billion-scale foundation model data.

Boundary precision: ASD decreased from 14–21 mm (2020 CNN methods) to 5.1–6.9 mm (2023–2024 transformer methods), indicating substantially improved boundary adherence beyond what Dice captures.

nnU-Net baseline: The self-configuring nnU-Net (tumor DC 0.810) outperforms numerous manually designed architectures published after it, demonstrating that systematic preprocessing and hyperparameter optimization are as important as architectural novelty.

Table III summarizes the loss functions and their properties relevant to liver tumor segmentation.

TABLE III
LOSS FUNCTIONS USED IN LIVER TUMOR SEGMENTATION

Loss Function	Imbalance Robust	Best Use Case
Binary Cross-Entropy	No	Balanced classes, calibration
Dice Loss	Yes	Primary overlap optimization
Focal Loss [6]	Yes	Extreme imbalance, hard examples
Tversky Loss	Yes	High recall for small lesions
Boundary Loss	Yes	Thin/irregular boundaries
Dice + Focal (comp.)	Yes	Most practical applications

Table IV reports cross-dataset generalization for key methods trained on LiTS and evaluated on 3DIRCADb.

TABLE IV
CROSS-DATASET GENERALIZATION (LITS → 3DIRCADB). PERFORMANCE DROP REFLECTS DOMAIN SHIFT BETWEEN INSTITUTIONS.

Method	LiTS DC	3DIRCADb DC	Drop
RA-UNet [15]	0.741	0.680	-8.2%
ResUNet [10]	0.792	0.734	-7.3%
AIM-Unet [12]	0.800	0.754	-5.8%
nnU-Net [11]	0.810	0.771	-4.8%
FUF-TransUNet [23]	0.834	0.898†	—
G-UNETR++ [24]	0.844	0.832	-1.4%

† Trained and tested independently on 3DIRCADb; not directly comparable.

Domain shift causes 5–8% tumor Dice performance drops for most methods. G-UNETR++ shows the smallest drop (-1.4%), suggesting that gradient-enhanced representations encode more scanner-agnostic boundary features. nnU-Net also generalizes well (-4.8%) due to automatic preprocessing adaptation to target dataset statistics.

VI. OPEN CHALLENGES AND FUTURE DIRECTIONS

A. Open Challenges

Small Tumor Detection: Detection and segmentation of lesions < 1 cm remain far below clinical standards; most methods achieve sensitivity < 0.50 for sub-centimeter lesions. This is the most critical gap for early HCC surveillance where curative treatment (resection or ablation) is indicated at the 1 cm threshold.

Class Imbalance: Tumor voxels constitute $< 3\%$ of total CT volume in LiTS (median 0.8%), with early-stage lesions at $< 0.1\%$. While compound Dice + Focal losses and tumor-biased patch sampling partially mitigate this, no current approach achieves adequate sensitivity for very small lesions without sacrificing specificity.

Domain Shift: Performance drops of 5–8% when transferring across institutions, scanner models, or acquisition protocols remain clinically problematic. Instance normalization, adversarial domain adaptation, and test-time augmentation provide partial mitigation.

Annotation Scarcity: Expert pixel-wise annotation of 3D CT volumes requires 20–60 minutes per case. The LiTS training set (131 volumes) is orders of magnitude smaller than natural image datasets used to pre-train ViT models. Semi-supervised and self-supervised frameworks are essential for clinical-scale deployment.

Prospective Clinical Validation: Retrospective benchmark performance does not guarantee clinical utility. No reviewed study provides prospective workflow integration, regulatory compliance assessment, or clinical decision impact evaluation.

Computational Cost: 3D transformer models require 40–120 GB GPU VRAM for full-volume inference, incompatible with standard clinical workstations (10–24 GB VRAM). Patch-based inference extends latency to 2–10 minutes per CT volume, insufficient for real-time applications.

B. Future Research Directions

Foundation Model Fine-Tuning: Systematic adaptation of SAM, MedSAM, and SAM-Med3D [37] to CT liver tumor segmentation using adapter fine-tuning (LoRA), task-specific prompt engineering, and domain-specific masked CT modeling pretraining could achieve strong performance with < 50 annotated cases.

Federated and Privacy-Preserving Learning: Multi-institutional FL frameworks (FedLiverNet [29]) with differential privacy guarantees and personalized local adaptation (IOP-FL) can aggregate clinical datasets across GDPR/HIPAA constraints while improving demographic diversity and generalization.

Self-Supervised Pretraining on Clinical CT Archives: Hospital PACS systems contain millions of unlabeled CT studies. Masked image modeling objectives applied to these corpora could provide powerful initialization for tumor segmentation with dramatically reduced labeling requirements—analogous to the impact of BERT on NLP.

Uncertainty Quantification: Bayesian neural networks, Monte Carlo Dropout, deep ensembles, and evidential deep learning provide per-voxel confidence maps, enabling human-AI collaborative workflows where uncertain regions are flagged for radiologist review.

Multi-Phase and Multi-Modal Fusion: Integrating arterial, portal-venous, and delayed CT phases within a unified segmentation framework, potentially with MRI fusion, can capture complementary diagnostic information unavailable in single-phase CT.

Lightweight Real-Time Architectures: MobileNetV2/EfficientNet encoder-based U-Nets, knowledge

distillation from large teacher models, and INT8/FP16 quantization can reduce inference latency to $< 100\,\text{ms}$ for intraoperative real-time applications.

VII. Conclusion

This paper presented a comprehensive survey and comparative analysis of deep learning methods for liver tumor segmentation from CT and MRI, covering 20+ architectures across five methodological paradigms over 2020–2024. The field has progressed through CNN-based encoder-decoder networks, attention-augmented U-Net variants, and Vision Transformer hybrids to foundation model adaptations.

Key findings are: (1) Liver segmentation Dice (> 0.96) is near-saturated on LiTS; tumor segmentation remains the dominant challenge; (2) Transformer-based architectures provide consistent improvements for boundary-adjacent and small tumors; (3) Two-stage pipeline decomposition (liver localization $\rightarrow$ tumor segmentation) consistently outperforms single-stage methods by 5–10%; (4) Compound Dice + Focal loss functions outperform single-loss training under class imbalance; (5) nnU-Net's self-configuration demonstrates that preprocessing and training protocol optimization are as impactful as architectural novelty; (6) Domain shift (5–8% Dice drop across institutions) remains the primary barrier to clinical translation.

State-of-the-art tumor Dice scores of 0.834 (FUF-TransUNet), 0.844 (G-UNETR++), and 0.943 (SAM + GAN) on LiTS represent remarkable progress but fall short of clinical-grade reliability for small ($< 1\,\text{cm}$) and infiltrative tumors. Future work must prioritize small tumor detection, domain robustness, federated multi-institutional training, and prospective clinical validation to translate benchmark performance into patient benefit.

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References

[1] F. Bray, M. Laversanne, H. Sung, J. Ferlay, R. L. Siegel, I. Soerjomataram, and A. Jemal, "Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries," *CA Cancer J. Clin.*, vol. 74, no. 3, pp. 229–263, 2024.

[2] E. A. Eisenhauer et al., "New response evaluation criteria in solid tumours: revised RECIST guideline (version 1.1)," *Eur. J. Cancer*, vol. 45, no. 2, pp. 228–247, 2009.

[3] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. MICCAI*, LNCS vol. 9351, pp. 234–241, Springer, 2015.

[4] P. Bilic et al., "The liver tumor segmentation benchmark (LiTS)," *Med. Image Anal.*, vol. 84, p. 102680, Feb. 2023.

[5] R. M. Devi and V. Seenivasagam, "Automatic segmentation and classification of liver tumor from CT image using feature difference and SVM-based classifier–soft computing technique," *Soft Comput.*, vol. 24, no. 24, pp. 18591–18598, 2020.

[6] T. Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE/CVF ICCV*, pp. 2980–2988, 2017.

[7] F. Milletari, N. Navab, and S. A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. IEEE 3DV*, pp. 565–571, 2016.

[8] X. Li, H. Chen, X. Qi, Q. Dou, C. W. Fu, and P. A. Heng, "H-DenseUNet: Hybrid densely connected UNet for liver and tumor segmentation from CT volumes," *IEEE Trans. Med. Imag.*, vol. 37, no. 12, pp. 2663–2674, Dec. 2018.

[9] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-Net architecture for medical image segmentation," in *Proc. DLMIA/ML-CDS (MICCAI Workshop)*, LNCS vol. 11045, pp. 3–11, Springer, 2018.

[10] H. Rahman, A. Bukht, A. Imran, A. Tariq, S. E. Tu, and S. Alasmari, "A deep learning approach for liver and tumor segmentation in CT images using ResUNet," *Bioengineering*, vol. 9, no. 8, p. 368, 2022.

[11] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation," *Nature Methods*, vol. 18, pp. 203–211, 2021.

[12] F. Özcan, O. N. Uçan, S. Karaçam, and D. Tunçman, "Fully automatic liver and tumor segmentation from CT image using an AIM-Unet," *Bioengineering*, vol. 10, no. 2, p. 215, Feb. 2023.

[13] J. Li et al., "EREs-UNet++: Liver CT image segmentation based on high-efficiency channel attention and Res-UNet++," *Comput. Biol. Med.*, vol. 158, p. 106501, May 2023.

[14] O. Oktay et al., "Attention U-Net: Learning where to look for the pancreas," in *Proc. MIDL*, 2018.

[15] Q. Jin, Z. Meng, C. Sun, H. Cui, and R. Su, "RA-UNet: A hybrid deep attention-aware network to extract liver and tumor in CT scans," *Front. Bioeng. Biotechnol.*, vol. 8, p. 1471, 2020.

[16] L. Jiang, J. Ou, R. Liu, Y. Zou, T. Xie, H. Xiao, and T. Bai, "RMAU-Net: Residual multi-scale attention U-Net for liver and tumor segmentation in CT images," *Comput. Biol. Med.*, vol. 158, p. 106838, 2023.

[17] X. Wang, S. Wang, Z. Zhang, X. Yin, T. Wang, and N. Li, "CPAD-Net: Contextual parallel attention and dilated network for liver tumor segmentation," *Biomed. Signal Process. Control*, vol. 79, Jan. 2023.

[18] S. Saumiya and S. W. Franklin, "Unified automated deep learning framework for segmentation and classification of liver tumors," *J. Supercomput.*, vol. 80, pp. 2347–2380, 2024.

[19] A. Hatamizadeh, Y. Tang, V. Nath, D. Yang, A. Myronenko, B. Landman, H. R. Roth, and D. Xu, "UNETR: Transformers for 3D medical image segmentation," in *Proc. IEEE/CVF WACV*, pp. 574–584, 2022.

[20] H. Cao, Y. Wang, J. Chen, D. Jiang, X. Zhang, Q. Tian, and M. Wang, "Swin-UNet: Unet-like pure transformer for medical image segmentation," in *Proc. ECCV*, pp. 205–218, Springer, 2022.

[21] H. Y. Zhou, J. Guo, Y. Zhang, X. Han, L. Wang, and Y. Yu, "nnFormer: Volumetric medical image segmentation via a 3D transformer," *IEEE Trans. Image Process.*, vol. 32, pp. 4036–4045, 2023.

[22] S. Di, Y. Zhao, M. Liao, F. Zhang, and X. Li, "TD-Net: A hybrid end-to-end network for automatic liver tumor segmentation from CT images," *IEEE J. Biomed. Health Inform.*, vol. 27, pp. 1163–1172, 2023.

[23] L. Song et al., "FUF-TransUNet: A transformer-based U-Net with fully utilize of features for liver and liver-tumor segmentation in CT images," in *Proc. Pattern Recognit. Comput. Vis. (PRCV)*, Springer, 2024.

[24] H. Wu et al., "Deep learning-based liver tumor segmentation from computed tomography scans with a gradient-enhanced network," *PMC – MDPI Sensors*, 2026.

[25] K. Guo, J. Chen, T. Qiu et al., "MedGAN: An adaptive GAN approach for medical image generation," *Comput. Biol. Med.*, vol. 163, p. 107119, 2023.

[26] J. Liao, H. Wang, H. Gu, and Y. Cai, "Liver tumor segmentation method combining multi-axis attention and conditional generative adversarial networks," *PLOS ONE*, vol. 19, no. 12, p. e0312105, Dec. 2024.

[27] Q. Hu, Y. Chen, J. Xiao et al., "Label-free liver tumor segmentation," in *Proc. IEEE/CVF CVPR*, pp. 7422–7432, 2023.

[28] "Foundation model-guided multi-view semi-supervised CT segmentation of liver tumors in resource-constrained settings," *npj Digit. Med.*, Nature, 2025, doi: 10.1038/s41746-025-02190-0.

[29] "FedLiverNet: A federated learning framework for privacy-preserving and efficient liver cancer detection," *Sci. Rep.*, Nature, 2025, doi: 10.1038/s41598-026-46244-z.

[30] U. Demir et al., "Liver tumor segmentation using SAM combining transformer with GANs," in *Proc. IEEE ISBI*, 2023.

[31] L. Soler, A. Hostettler, A. Charnoz, J. Fasquel, and J. Moreau, "3D image reconstruction for comparison of algorithm database: A patient-

specific anatomical and medical image database," IRCAD, Strasbourg, France, Tech. Rep., 2010.

[32] A. E. Kavur et al., "CHAOS challenge – combined (CT–MR) healthy abdominal organ segmentation," *Med. Image Anal.*, vol. 69, p. 101950, 2021.

[33] P. F. Christ et al., "Automatic liver and tumor segmentation of CT and MRI volumes using cascaded fully convolutional neural networks," *arXiv preprint arXiv:1702.05970*, 2017.

[34] S. T. Tran, C. H. Cheng, and D. G. Liu, "A multiple layer U-Net, Un-Net, for liver and liver tumor segmentation in CT," *IEEE Access*, vol. 9, pp. 3752–3764, 2020.

[35] S. Almotairi, G. Kareem, M. Aouf, B. Almutairi, and M. A.-M. Salem, "Liver tumor segmentation in CT scans using modified SegNet," *Sensors*, vol. 20, no. 5, p. 1516, Mar. 2020.

[36] C. Zhang, J. Lu, Q. Hua, C. Li, and P. Wang, "SAA-Net: U shaped network with scale-axis-attention for liver tumor segmentation," *Biomed. Signal Process. Control*, vol. 73, p. 103460, 2022.

[37] H. Wang et al., "SAM-Med3D: Towards general-purpose segmentation models for volumetric medical images," in *Proc. ECCV*, pp. 51–67, Springer, 2024.